

Competitive Exam Practice 10 — Answer Key

Virtual Memory and Secondary Storage

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Provenance is repeated for auditability: Q1 is a verified real PYQ; Q2–Q8 are original GATE-pattern questions. The derivations are independent of the question file and were checked arithmetically after drafting.

- Q1** [*GATE CS 2004 Q47, verified real PYQ — GATEOverflow V3 p.92*] **1260 ns.** A TLB hit costs 150 ns for the memory access. A TLB miss costs $3 \times 150 = 450$ ns because two page-table accesses and the final access are required. Thus the mean cost per access is $0.9(150) + 0.1(450) = 180$ ns, and two accesses cost 360 ns. The mean page-fault cost is $8 \text{ ms}/10,000 = 800$ ns per instruction. Therefore $100 + 360 + 800 = 1260$ ns.
- Q2** [*Original, GATE-pattern*] **9438.** A 1 KiB page has a 10-bit offset. For 0xABCDE, the low ten bits are $0x0DE = 222$ and the VPN is the high ten bits, 687. With frame 9, the physical address is $9 \times 1024 + 222 = 9438$ (0x24DE). The offset is unchanged.
- Q3** [*Original, GATE-pattern*] **70 ns.** A hit costs $10 + 50 = 60$ ns. A miss costs $10 + 3(50) = 160$ ns. Hence $0.9(60) + 0.1(160) = 54 + 16 = 70$ ns.
- Q4** [*Original, GATE-pattern*] **(B).** The offset must satisfy $0 \leq 550 < 600$, so the reference is valid. Segmentation then forms $\text{base} + \text{offset} = 12,000 + 550 = 12,550$.
- Q5** [*Original, GATE-pattern*] **88 KiB.** Best-fit chooses the smallest hole that can hold 212 KiB, namely 300 KiB. The selected remainder is $300 - 212 = 88$ KiB.
- Q6** [*Original, GATE-pattern*] **1011.** $1101 \oplus 0011 = 1110$ and $1110 \oplus 0101 = 1011$, so $P = 1011$.
- Q7** [*Original, GATE-pattern*] **8.4667 ms (approximately).** At 7200 rpm, one rotation takes $60/7200 = 8.3333$ ms, so average rotational latency is 4.1667 ms. The total is $4 + 4.1667 + 0.2 + 0.1 = 8.4667$ ms.
- Q8** [*Original, GATE-pattern*] **(C).** RAID 0 uses striping without redundancy. Paging uses fixed-size frames and can leave internal fragmentation; segmentation uses variable-size logical units; RAID 1 uses mirroring, while RAID 6 uses dual parity.